

MacroWatch README

Architecture, File Structure & System Overview

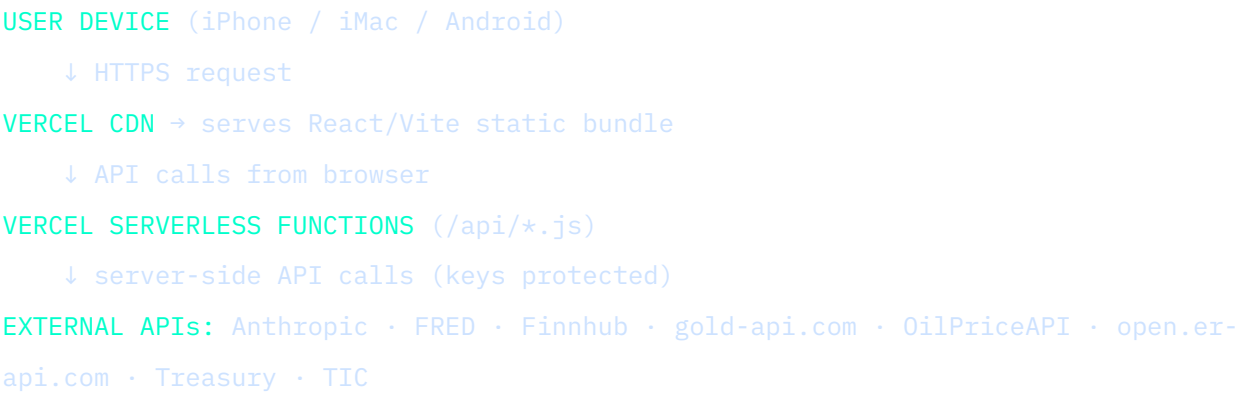
v1.1-stable | Live: <https://macro-watch-phi.vercel.app> | GitHub:
<https://github.com/mlibrizzi/MacroWatch>

What is MacroWatch?

MacroWatch is a personal macro intelligence dashboard for monitoring US debt cycle dynamics through Ray Dalio's Big Debt Cycle framework. It aggregates data from 8+ APIs into a unified mobile-first PWA (Progressive Web App) deployed on Vercel and installable on iPhone, Android, and desktop.

The dashboard answers one central question: **Where are we in the long-term debt cycle, and what does that mean for capital preservation?**

System Architecture



Component Roles

Component	Role	Details
Anthropic Claude	AI narrative generation	Powers Intel briefing, Fed narrative, government report summaries. Model: claude-haiku-4-5-20251001. max_tokens: 4000. Proxied server-side via api/claude.js.
GitHub	Source code	Repo: mlibrizzi/MacroWatch. All code changes pushed via GitHub REST API from iMac Terminal using Personal Access Token. Auto-triggers

	repository	Vercel deploy on every push to main.
Vercel	Hosting + serverless functions	Free hobby tier. Auto-deploys on GitHub push. Serves static React bundle + /api/* serverless functions. Stores all API keys as encrypted environment variables.
React/Vite	Frontend framework	React 18, Vite build tool, Recharts for charts. All UI in App.jsx (~970 lines). Deployed as static bundle with PWA support (installable on iPhone home screen).
FRED API	Macro data backbone	Treasury yields, CPI, PCE, GDP, VIX (VIXCLS), Fed balance sheet (H.4.1), TGA, WTI/Brent historical. St. Louis Federal Reserve. Free, official government data.
Finnhub	Market data	Real-time Mag7 stock prices, SPX/NDX/DJI via ETF proxies (SPY×10, QQQ×29, DIA×100), quarterly earnings vs estimates. Free tier. 60 req/min.
gold-api.com	Precious metals	Real-time gold (XAU/USD) and silver (XAG/USD) spot prices.
OilPriceAPI.com	Oil prices	Near real-time WTI and Brent crude (~30 min refresh). Free tier, 1000 req/month. Selected after FRED 1-week lag caused missed -10% oil move (May 6, 2026).
Treasury Fiscal Data	Auction results	Official same-day auction data: BTC, indirect bidders %, dealer %, high yield. Free, no key required. api.fiscaldata.treasury.gov
Treasury TIC	Foreign holdings	slt_table5.txt — major foreign holders of US Treasuries. Official, free, 45-day lag. ticdata.treasury.gov
open.er-api.com	FX rates	EUR/USD, USD/JPY, GBP/USD. Free, hourly update. No key required.

File Structure

Root Directory

File / Folder	Purpose
<code>package.json</code>	Node.js dependencies and build scripts (npm run build → vite build)
<code>vite.config.js</code>	Vite build configuration — React plugin only. Minimal config.
<code>index.html</code>	HTML entry point for the React app
<code>/api/</code>	Vercel serverless functions — all external API calls

/src/	React frontend source code
/docs/	Documentation PDFs and markdown files
/public/	Static assets (favicon, PWA manifest if present)

/api — Vercel Serverless Functions

File	Purpose	External API
claude.js	Anthropic API proxy. Adds system prompt enforcement (forces raw JSON, strips markdown fences). All Claude calls go through this server-side to protect API key.	api.anthropic.com
fred.js	Fetches all FRED data: yields (2Y/10Y/30Y/TIPS/FEDFUNDS), inflation (CPI/PCE/Core), labor (unemployment/NFP/claims), GDP, retail sales, Fed balance sheet (WALCL/TREAST/WSHOMCB/WRBWFRBL), TGA (WDTGAL), VIX (VIXCLS), WTI/Brent historical	api.stlouisfed.org
markets.js	Fetches Finnhub: Mag7 stock quotes, ETF quotes for index proxies (SPY/QQQ/DIA), FX rates from open.er-api.com	finnhub.io, open.er-api.com
metals.js	Real-time gold and silver spot prices	gold-api.com
oil.js	Near real-time WTI (WTI_USD) and Brent (BRENT_CRUDE_USD) with 24h change. Added May 8, 2026 to fix FRED oil lag issue.	api.oilpriceapi.com
auctions.js	Fetches Treasury auction results. Calculates WEAK/MIXED/STRONG/PENDING status based on BTC, indirect%, dealer% thresholds. Sorts results by maturity descending (30Y→2Y).	api.fiscaldata.treasury.gov
earnings.js	Quarterly earnings for AAPL/MSFT/NVDA/GOOGL/AMZN/META/TSLA/JPM/XOM. EPS actual vs estimate, beat/miss %, verified against SEC EDGAR.	finnhub.io
tic.js	Parses Treasury TIC slt_table5.txt (tab-delimited). Returns top 8 foreign holders with MoM change and buying/selling trend direction.	ticdata.treasury.gov

/src — React Frontend Source

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File	Purpose & Details
<code>main.jsx</code>	React app entry point. Mounts App component to DOM element #root.
<code>App.jsx</code>	Main application file (~970 lines). Contains ALL UI components: tab navigation, DailyTab, AuctionsTab, WeeklyTab, MonthlyTab, QuarterlyTab, IntelTab, Loading spinner, RegimeSignals, and all CSS styles. Single-file architecture for simplicity.
<code>fetchers.js</code>	Data orchestration layer (~370 lines). Contains: <code>fetchLive()</code> (all API calls with cache-busting), <code>fetchDaily()</code> , <code>fetchWeekly()</code> , <code>fetchMonthly()</code> , <code>fetchQuarterly()</code> , <code>fetchIntel()</code> . All tab data flows through here.
<code>api.js</code>	Anthropic Claude helper. <code>callClaude(prompt, systemOverride)</code> function. Sets <code>max_tokens:4000</code> , <code>model</code> , handles JSON parsing with fallback to <code>{raw: text}</code> to prevent crashes on malformed responses.

Environment Variables

All API keys stored as Vercel environment variables (Settings → Environment Variables). Never committed to GitHub. Access in serverless functions via `process.env.KEY_NAME`. In React frontend via `import.meta.env.KEY_NAME` (VITE_ prefix required for Vite).

Variable Name	Used In	Where to Get It
<code>VITE_ANTHROPIC_API_KEY</code>	<code>api/claude.js</code> , <code>src/api.js</code>	console.anthropic.com → API Keys
<code>VITE_FRED_API_KEY</code>	<code>api/fred.js</code>	fred.stlouisfed.org → My Account → API Keys
<code>VITE_FINNHUB_API_KEY</code>	<code>api/markets.js</code> , <code>api/earnings.js</code>	finnhub.io → Dashboard
<code>VITE_GOLD_API_KEY</code>	<code>api/metals.js</code>	gold-api.com → Dashboard
<code>VITE_OILPRICE_API_KEY</code>	<code>api/oil.js</code>	oilpriceapi.com → Dashboard

How to Make Code Changes

All code changes are made via GitHub REST API from iMac Terminal using a Personal Access Token. Never paste code directly into GitHub web UI — always use Python scripts via the heredoc pattern.

Standard Workflow

Step 1: export GH_TOKEN='your_token_here'

Step 2: Fetch current file:
urllib.request.urlopen('https://raw.githubusercontent.com/mlibrizzi/MacroWatch/main/...')

Step 3: Make surgical replacement: code.replace(old_string, new_string)

Step 4: Verify change applied + check smart quotes count = 0

Step 5: Upload via GitHub API PUT with base64 encoded content + file SHA

Step 6: Wait for Vercel auto-deploy (15-20 seconds typical)

Step 7: Test via curl or browser before declaring success

GitHub Personal Access Token

Generate at: [github.com](#) → **Settings** → **Developer settings** → **Personal access tokens** → **Tokens (classic)** → **Generate new token**

Required scope: **repo** (full control of private repositories). Token expires based on settings — re-generate when expired. A 401 Unauthorized error in Terminal = expired token.

Stable Version Tags

Tag	Date	Description
v1.0-stable	May 5, 2026	First fully verified baseline. All tabs showing real data. Intel working with hardcoded anchors.
v1.1-stable	May 8, 2026	Real-time oil (OilPriceAPI), cache busting for all API calls, auction improvements (PENDING status, maturity sort, hidden undefined fields).

Hardcoded Values – Manual Update Required

 **These values do NOT update automatically.** They are fixed text strings in the source code. When the underlying data changes in the real world, the app continues showing the old value until the code is manually edited and redeployed to Vercel.

Field	Current Value	Location	Update When
Fed Funds Rate	3.50%-3.75%	src/fetchers.js → fetchWeekly	After each FOMC meeting (next: June 16-17, 2026)

June cut probability	6%	src/fetchers.js → fetchWeekly	Weekly — check CME FedWatch
FOMC dissents	4	src/fetchers.js → fetchWeekly	After each FOMC meeting
Next meeting date	2026-06-16	src/fetchers.js → fetchWeekly	After each FOMC meeting
Federal Deficit YTD	\$1.307T (Mar 2026)	src/fetchers.js → fetchQuarterly	Monthly — Treasury Monthly Statement
Federal Debt	\$38.9T	src/fetchers.js → fetchQuarterly	Monthly — TreasuryDirect
Debt-to-GDP	122%	src/fetchers.js → fetchQuarterly	Quarterly — BEA/CBO
Trade Balance	-\$60.3B (Mar 2026)	src/fetchers.js → fetchQuarterly	Monthly — BEA release
China net selling since 2021	-\$185B	api/tic.js	Monthly — after TIC release
Intel price anchors	Gold \$4718, 10Y 4.45%, VIX 18 etc.	src/fetchers.js → fetchIntel	Weekly — if price moves significantly